Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of OpenUSD-based Automated Digital Twin Construction and AI-driven Intelligent Operational Optimization for Buildings			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI (√)**, Next-gen Communication ( ), **Advanced Robotics & Manufacturing (√)**, Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Knowledge Services*	
	Medium Category	Small Category	Medium Category	Small Category
	Software	*Software Solutions*	*R&D / Engineering Services*	*Design Information Integration Management / Collaboration System Performance Enhancement Technologies*
Concept of Proposed Technology	This technology is an "AI-Integrated Automated Digital Twin Orchestration Platform" designed to overcome the high cost and long lead-time barriers of conventional digital twin construction. At its core is the Twin Bridge engine, which enables 1-click automated conversion of industrial standard data (Revit, CAD, BIM) into OpenUSD format, reducing the construction period by over 95% (from 6 months to within 4 weeks). Beyond visualization, it integrates Vision AI for real-time occupant behavior analysis and Synthetic Data for simulating edge-case scenarios, providing a Full Stack Digital Twin environment from data ingestion to AI-driven optimization.			

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Rationale for Joint R&D Project	The global digital twin market is currently facing significant barriers to widespread adoption due to the difficulty in securing specialized modeling talent, excessive initial PoC costs exceeding $400,000, and a lack of domain-specific algorithms that accurately reflect the complex physical characteristics of industrial sites. Conventional manual modeling processes, which often require several months just to establish a data pipeline, act as a primary bottleneck preventing enterprises from achieving agile digital transformation. To address these structural challenges, it is imperative to forge a strategic collaboration by integrating the participating Korean SME's NVIDIA Omniverse-based OpenUSD automated infrastructure and data conversion technology with Professor Seungjae Lee's core expertise in building physics simulation and occupant-centric interaction from the University of Toronto's Intelligent and Interactive Buildings Lab (IIB Lab). This international joint R&D project will ensure technical alignment with OpenUSD, the emerging global standard for 3D data, thereby completing a standard digital twin platform capable of penetrating the North American smart building and construction automation markets. This collaboration will lead to world-class academic validation through the University of Toronto's research and facilitate the global commercialization of innovative Korean technology, creating significant economic value and establishing a leading model for next-generation digital twin solutions.
Key Words	1. Automated 3D Data Integration Engine 2. Open Standard 3D Data Format 3. High-Efficiency Digital Twin Construction 4. Intelligent Human-Building Interaction 5. Virtual Environment-based AI Training

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (3, 4)	**End of the Project**	Level (7, 8)
R&D Objectives	• Streamlining construction processes by developing automated conversion technology for industrial design data into 3D standard formats • Advancing intelligent spatial monitoring and predictive simulation based on vision analytics and synthetic data • Commercializing an open-standard-based digital twin operational platform and validating global technical standards			
R&D Target Specifications	• Construction Duration: Over 95% reduction compared to manual processes (from 6 months to within 4 weeks) • Data Conversion Integrity: Over 99% success rate in translating design data into 3D standard formats • AI Analytics Accuracy: Over 90% detection accuracy for on-site objects using virtual environment-based training data			
R&D Plan	**Final R&D Goal** Development and global validation of a next-generation intelligent digital twin orchestration platform that integrates automated OpenUSD conversion technology for industrial data with vision AI-based interaction analysis, reducing conventional construction costs by over 80%.			

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Annual Objectives and Detailed Tasks

1st Year Objectives:

- Achieving a 95% reduction in digital twin construction time by automating the manual 3D modeling process.
- Establishing a high-performance data interoperability framework based on the OpenUSD standard for large-scale industrial assets.

1st Year Detailed Tasks:

- Researching and developing a "1-Click" automated conversion algorithm that translates industrial design formats (Revit, STEP, IGES) into OpenUSD schemas while preserving geometry and metadata.
- Implementing automated Level of Detail (LOD) generation and polygon optimization techniques to enable real-time rendering of complex factory and building models.
- Developing a metadata mapping engine that synchronizes design-stage information, such as materials and specifications, with the physics engine of the digital twin.

2nd Year Objectives:

- Implementing real-time "Sim-to-Real" synchronization between physical site data and the virtual environment.
- Improving the reliability of AI analysis models for construction and manufacturing domains using high-fidelity synthetic data.

2nd Year Detailed Tasks:

- Developing a synchronization module that maps real-time object movements and behaviors from CCTV feeds onto virtual characters within the digital twin using advanced vision AI.
- Building a synthetic data generation pipeline to simulate critical edge-case scenarios, such as structural failures or safety accidents, which are difficult to collect in real-world environments.
- Designing and integrating physics-based simulation algorithms that reflect the mechanical properties of building structures in collaboration with the research institute.

3rd Year Objectives:

- Completing an integrated orchestration platform with distributed computing optimization for global scalability.
- Validating technical efficacy and global standards through pilot tests at smart building and industrial sites in North America.

3rd Year Detailed Tasks:

- Optimizing distributed resource management and scheduling to support simultaneous multi-user access and large-scale computational tasks for massive digital twins.
- Developing an intelligent user interface and analytics dashboard integrated with Large Language Models (LLMs) for natural language-based twin control and reporting.
- Conducting on-site Proof of Concept (PoC) tests with North American partners to verify system reliability and obtaining international technical standard certifications.

Research Approach and Methodologies

1. Automated Data Pipeline and OpenUSD-based Interoperability

To overcome the limitations of traditional manual-centric modeling, we will establish an automated data pipeline based on the OpenUSD standard. Our approach focuses on developing a specialized conversion engine that analyzes the hierarchical structures and metadata of various industrial design files, such as Revit and CAD, ensuring high-fidelity migration to the virtual environment. Specifically,

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	through schema-mapping technology, we adopt a core methodology that enables the 1-Click synchronization of not only geometry but also metadata like materials and specifications, aiming for a 95% reduction in construction time. **2. Hybrid Vision AI and Synthetic Data Training Framework** To maximize the recognition rate of AI models in complex industrial environments, we will employ a hybrid training strategy that combines real-world data with synthetic data generated within virtual environments. While collecting field data through real-time vision analytics, we apply a methodology to generate large-scale "edge-case" scenarios, such as fire, structural failure, or equipment malfunction, which are difficult to capture in reality. Through this Sim-to-Real approach, we establish an AI training framework capable of ensuring high detection accuracy (90%+) even when data collection is restricted. **3. Physics-based Simulation and Structural Validation** Beyond simple visual implementation, we will apply a physics-based simulation methodology using the NVIDIA PhysX engine to ensure consistency between the virtual and physical worlds. By embedding mechanical properties and physical constraints of buildings into the digital twin, and through joint research with the research institute, we will perform Bayesian inference and causal modeling to calibrate errors between simulation results and real-world sensor data. This ensures the reliability required for the digital twin to contribute to structural health monitoring and energy optimization. **4. Scalable Cloud-Native Infrastructure and Agentic Interface** To support large-scale industrial deployment, we will apply a containerized microservices architecture and distributed GPU resource management methodologies. We optimize GPU scheduling for real-time computational processing of massive digital twin assets and establish an agentic interface integrated with Large Language Models (LLMs) and Vision Language Models (VLMs). We finalize a user-centric operational methodology where the system interprets natural language commands to execute simulations and automatically generate analytical reports.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development of core algorithms for building physics and occupant interaction • Design of structural health monitoring and energy optimization scenarios • Academic validation of simulation accuracy and Sim-to-Real consistency • Support for North American market verification and local data acquisition
Expected R&R of Korean SME Partner	• Development of the automated OpenUSD data conversion and integration engine • Implementation of the AI-driven digital twin orchestration platform infrastructure • Integration of real-time vision analytics and synthetic data generation modules • Productization, commercialization, and global business development

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements** • Securing global leadership in automated digital twin construction technology based on OpenUSD standard • Establishing a highly reliable AI training framework for industrial edge-case scenarios • Advancing physically accurate simulation technology for smart building management **Commercialization Potential** • Establishing a pipeline for global market expansion, including the North American construction and manufacturing sectors • Expanding the digital twin service model to Tier 1 industries such as semiconductors and bio-pharmaceuticals • Achieving sustainable revenue growth through standardized software packaging and SaaS models **Industrial and Economic Impact** • Lowering the barrier to entry for digital transformation by reducing PoC costs by over 80% • Improving industrial safety and reducing social costs through predictive maintenance and risk simulation • Accelerating the growth of the domestic digital twin ecosystem through global technical standard alignment